

Kernel L0 Dashboard

Mathematical and Accounting Specification

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Version 1.0 — Frozen Specification

Normative Status

This document is a **normative specification**. It defines the authoritative mathematical and accounting meaning of Kernel L0 Dashboard Tier 1 metrics.

Any implementation, test, or visualization that contradicts this document is incorrect by definition.

Specification Mode

Each metric is defined simultaneously in:

1. Formal Mathematical Notation
2. Accounting Narrative Language (GAAP-style)

Both representations are **equally authoritative**. If a discrepancy exists, the specification is invalid and must be revised via a delta document.

Scope and Constraints

- Accrual basis
- Kernel L0 accounts only
- Read-only (no journal entry creation)
- No projections, recommendations, or scoring
- Period-aware

1 Preliminaries

1.1 Fiscal Period Model

Let p denote a fiscal period selected by the user (e.g., month, quarter, year).

Let p^- denote the immediately preceding fiscal period.

1.2 Measurement Functions

Because balance sheet accounts are *stock* measures and income statement accounts are *flow* measures, define:

- Period-end balance function:

$$B(a, p) \in \mathbb{R}$$

representing the balance (in reporting polarity) of account a at the close of period p .

- Period activity function:

$$T(a, p) \in \mathbb{R}$$

representing the total activity (in reporting polarity) of income statement account a over period p .

1.3 Delta Operator

For any metric $M(p)$:

$$\Delta M(p) = M(p) - M(p^-)$$

If either term is undefined, $\Delta M(p)$ is undefined.

2 Kernel L0 Account Sets

2.1 Cash

$$\mathcal{A}_{\text{cash}} = \{10000, 10100\}$$

2.2 Revenue and Refunds

$$\mathcal{A}_{\text{rev}} = \{40000\} \quad \mathcal{A}_{\text{ref}} = \{49000\}$$

2.3 Operating Costs

$$\mathcal{A}_{\text{cogs}} = \{50000\}$$

$$\mathcal{A}_{\text{opex}} = \{60000\}$$

$$\mathcal{A}_{\text{pay}} = \{61000\}$$

$$\mathcal{A}_{\text{dep}} = \{62000\}$$

2.4 Current Assets (Kernel Approximation)

$$\mathcal{A}_{\text{CA}} = \{10000, 10100, 12000, 14000\}$$

2.5 Current Liabilities (Kernel Approximation)

$$\mathcal{A}_{\text{CL}} = \{20000, 21000, 22000, 23000\}$$

3 Metric Specifications

3.1 Metric: Total Cash

Formal Mathematical Definition

$$\text{TotalCash}(p) = \sum_{a \in \mathcal{A}_{\text{cash}}} B(a, p)$$

Domain:

$$\text{TotalCash}(p) \in \mathbb{R}$$

Accounting Narrative Definition

Total Cash represents immediate liquidity available to the entity, comprising operating cash and undeposited funds. It is derived from period-end balances.

Undefined Cases

None.

Audit Rule

Reproducible from the trial balance without estimates.

3.2 Metric: Net Revenue

Formal Mathematical Definition

$$\text{NetRevenue}(p) = \sum_{a \in \mathcal{A}_{\text{rev}}} T(a, p) - \sum_{a \in \mathcal{A}_{\text{ref}}} T(a, p)$$

Accounting Narrative Definition

Net Revenue represents recognized revenue reduced by refunds and allowances for the selected period.

Undefined Cases

None.

Audit Rule

Derived solely from income statement totals.

3.3 Metric: Operating Income

Formal Mathematical Definition

Let:

$$\text{OperatingCosts}(p) = \sum_{a \in \mathcal{A}_{\text{cogs}} \cup \mathcal{A}_{\text{opex}} \cup \mathcal{A}_{\text{pay}} \cup \mathcal{A}_{\text{dep}}} T(a, p)$$

Then:

$$\text{OperatingIncome}(p) = \text{NetRevenue}(p) - \text{OperatingCosts}(p)$$

Accounting Narrative Definition

Operating Income represents profit from core operations before taxes, interest, and financing effects.

Explicit Exclusions

Taxes, interest, equity effects, financing activities.

Undefined Cases

None.

3.4 Metric: Net Working Capital

Formal Mathematical Definition

$$\text{CurrentAssets}(p) = \sum_{a \in \mathcal{A}_{CA}} B(a, p)$$

$$\text{CurrentLiabilities}(p) = \sum_{a \in \mathcal{A}_{CL}} B(a, p)$$

$$\text{NWC}(p) = \text{CurrentAssets}(p) - \text{CurrentLiabilities}(p)$$

Accounting Narrative Definition

Net Working Capital represents the excess of current assets over current liabilities using the Kernel-defined approximation.

Undefined Cases

None.

3.5 Metric: Current Ratio

Formal Mathematical Definition

$$\text{CurrentRatio}(p) = \begin{cases} \frac{\text{CurrentAssets}(p)}{\text{CurrentLiabilities}(p)}, & \text{if } \text{CurrentLiabilities}(p) \neq 0 \\ \text{undefined}, & \text{if } \text{CurrentLiabilities}(p) = 0 \end{cases}$$

Accounting Narrative Definition

The Current Ratio is a liquidity indicator expressing current assets relative to current liabilities.

Undefined Cases

Division by zero; rendered as N/A.

Freeze Declaration

This specification is frozen for Kernel Dashboard v1.0. Any modification requires a delta document and version increment.

A Worked Numerical Examples

This appendix provides concrete numerical instantiations of the formal definitions in Section 3. All examples are authoritative and must reconcile exactly in any implementation.

All balances and totals are expressed in reporting polarity.

A.1 Example A.1 — Empty Company (Baseline)

Inputs (Period p) For all Kernel L0 accounts a :

$$B(a, p) = 0 \quad \text{and} \quad T(a, p) = 0$$

Results

$$\text{TotalCash}(p) = 0$$

$$\text{NetRevenue}(p) = 0$$

$$\text{OperatingIncome}(p) = 0$$

$$\text{NWC}(p) = 0$$

$$\text{CurrentRatio}(p) = \text{undefined}$$

Interpretation Constraint Zero is a valid value. Undefined ratios must be rendered as N/A .

A.2 Example A.2 — Cash Only

Inputs

$$B(10000, p) = 10,000$$

All other balances and totals are zero.

Results

$$\text{TotalCash}(p) = 10,000$$

$$\text{NWC}(p) = 10,000$$

$$\text{CurrentRatio}(p) = \text{undefined}$$

A.3 Example A.3 — Revenue with Refunds

Inputs

$$T(40000, p) = 50,000 \quad T(49000, p) = 8,000$$

Results

$$\text{NetRevenue}(p) = 50,000 - 8,000 = 42,000$$

$$\text{OperatingIncome}(p) = 42,000$$

A.4 Example A.4 — Full Operating Stack (Profit)

Inputs

$$T(40000, p) = 100,000$$

$$T(49000, p) = 5,000$$

$$T(50000, p) = 40,000$$

$$T(60000, p) = 20,000$$

$$T(61000, p) = 25,000$$

$$T(62000, p) = 3,000$$

Intermediate

$$\text{NetRevenue}(p) = 95,000$$

$$\text{OperatingCosts}(p) = 88,000$$

Results

$$\text{OperatingIncome}(p) = 7,000$$

A.5 Example A.5 — Net Working Capital (Positive)

Inputs — Assets

$$B(10000, p) = 20,000$$

$$B(10100, p) = 500$$

$$B(12000, p) = 12,000$$

$$B(14000, p) = 1,500$$

Inputs — Liabilities

$$B(20000, p) = 8,000$$

$$B(21000, p) = 2,000$$

$$B(22000, p) = 3,000$$

$$B(23000, p) = 6,000$$

Intermediate

$$\text{CurrentAssets}(p) = 34,000$$

$$\text{CurrentLiabilities}(p) = 19,000$$

Results

$$\text{NWC}(p) = 15,000$$

$$\text{CurrentRatio}(p) = \frac{34,000}{19,000} \approx 1.78947$$

A.6 Example A.6 — Division by Zero (Ratio Undefined)

Inputs

$$\text{CurrentAssets}(p) = 5,000 \quad \text{CurrentLiabilities}(p) = 0$$

Results

$$\text{NWC}(p) = 5,000$$

$$\text{CurrentRatio}(p) = \text{undefined}$$

Rendering Rule Undefined ratios must not be coerced into numeric values.

A.7 Example A.7 — Delta Computation

Inputs

$$\text{TotalCash}(p^-) = 9,000 \quad \text{TotalCash}(p) = 13,700$$

Result

$$\Delta \text{TotalCash}(p) = 13,700 - 9,000 = 4,700$$

Delta Constraint If either operand is undefined, the delta is undefined.

B Audit Reconciliation Procedure

For any metric defined in this specification, an independent reviewer must be able to:

1. Extract the relevant account balances and totals from the trial balance.
2. Identify the applicable account sets defined in Section 2.
3. Apply the formal mathematical definitions exactly as written.
4. Reconcile the computed value to the dashboard output.

Failure at any step indicates non-compliance with this specification.